MATH329 · OPERATIONS RESEARCH

# Lecture 3 Standard Form

Midterm summary · Chapter 3 · the algebraic solution

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## 01 Why do we need the standard form?

The graph works for two variables only. With more variables we solve algebraically with the **Simplex** method, which moves from one corner to a better corner until it reaches the optimum.

> The Simplex Method attempts to move from one corner point of the solution space to a better corner point until the optimum is found.

Before simplex starts, the problem must be written in the **Standard Form** (also called **Equation form**). This is a fixed question in every midterm, worth 5 to 8 marks.

**Appeared in the midterm (461)**

The question: what does standard form mean? The expected answer, word for word:

> 1. All decision variables are **nonnegative**.
> 2. All constraints (with the exception of the non-negativity of the variables) are **equations with nonnegative RHS**.

## 02 The conversion rules

| Case | What to do | Example |
|---|---|---|
| A $\leq$ constraint | Add a **slack variable** with a plus sign. | $x_1 + 3x_2 \leq 6$<br>$x_1 + 3x_2 + s_1 = 6$ |
| A $\geq$ constraint | Subtract a **surplus variable**, with a minus sign. | $3x_1 + 2x_2 \geq 6$<br>$3x_1 + 2x_2 - s_2 = 6$ |
| An $=$ constraint | It stays as it is, with no new variable. | $x_1 + x_2 = 5$ |
| A negative right-hand side | Multiply the whole constraint by $-1$. If it is an inequality, its sign flips. | $3x_1 + 2x_2 \geq -2$<br>$-3x_1 - 2x_2 \leq 2$ |
| A variable **unrestricted in sign** | Replace it everywhere, including the objective, by the difference of two nonnegative variables. | $x_2 = x_2^+ - x_2^-$ |
| The objective function | Add the new variables with zero coefficients. | $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$ |
| The last line | All variables, old and new, are $\geq 0$. | $x_1, x_2, s_1, s_2 \geq 0$ |

**The order matters**

First fix the negative right-hand side, then substitute the unrestricted variable, and add the slack and surplus variables last. Adding a slack before multiplying by $-1$ gives it the wrong sign.

## 03 The slide examples

### 1 SLACK Example 3.1

| |
|---|
| Maximize $z = 2x_1 + 3x_2$ |
| subject to $x_1 + 3x_2 \le 6$ |
| $3x_1 + 2x_2 \le 6$ |
| $x_1, x_2 \ge 0$ |

| |
|---|
| Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$ |
| subject to $x_1 + 3x_2 + s_1 = 6$ |
| $3x_1 + 2x_2 + s_2 = 6$ |
| $x_1, x_2, s_1, s_2 \ge 0$ |

### 2 SURPLUS The second constraint is now ≥

| | |
|---|---|
| Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$ | |
| subject to $x_1 + 3x_2 + s_1 = 6$ | $s_1$ slack |
| $3x_1 + 2x_2 - s_2 = 6$ | $s_2$ surplus |
| $x_1, x_2, s_1, s_2 \ge 0$ | |

### 3 NEGATIVE RHS The second constraint is $3x_1 + 2x_2 \geq -2$

$$3x_1 + 2x_2 \geq -2 \quad \times (-1) \quad \rightarrow \quad -3x_1 - 2x_2 \leq 2$$

$$\begin{aligned} &\text{Maximize } z = 2x_1 + 3x_2 + 0s_1 + 0s_2 \\ &\text{subject to} \quad x_1 + 3x_2 + s_1 = 6 \\ &\qquad\qquad -3x_1 - 2x_2 + s_2 = 2 \\ &\qquad\qquad x_1, x_2, s_1, s_2 \geq 0 \end{aligned}$$

### 4 UNRESTRICTED $x_1$ is unrestricted

$$\begin{aligned} &\text{Maximize } z = x_1 + 3x_2 \\ &\text{subject to} \quad x_1 + x_2 \leq 2 \\ &\qquad\qquad -x_1 + x_2 \leq 4 \\ &\qquad\qquad x_1 \text{ unrestricted}, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} &\text{Maximize } z = x_1^+ - x_1^- + 3x_2 + 0s_1 + 0s_2 \\ &\text{subject to} \quad x_1^+ - x_1^- + x_2 + s_1 = 2 \\ &\qquad\qquad -x_1^+ + x_1^- + x_2 + s_2 = 4 \\ &\qquad\qquad x_1^+, x_1^-, x_2, s_1, s_2 \geq 0 \end{aligned}$$

## 04 The midterm question: every rule in one problem

This question appeared in the same shape in 461 and 462 with different numbers. This is

the 462 version (8 marks):

Maximum $Z = 8x_1 - 2x_2$

S.to $2x_1 + x_2 \leq 12 \cdot x_1 - 5x_2 \geq 6 \cdot 4x_2 = -7$

$x_1 \geq 0$, $x_2$ unrestricted in sign.

1. What methods can be applied to solve the given linear programming problem?
2. Write the problem in standard form (equation form). "Don't solve the problem"

## 1 The methods

`Graphical method and Algebraic method.`

The graphical method applies because there are two variables, and the algebraic method always applies.

## 2 The negative right-hand side

$4x_2 = -7 \quad \times (-1) \quad \rightarrow \quad -4x_2 = 7$

## 3 Substitute $x_2 = x_2^+ - x_2^-$ everywhere

$-2x_2 \quad \rightarrow \quad -2x_2^+ + 2x_2^-$

$x_2 \quad \rightarrow \quad x_2^+ - x_2^-$

$-5x_2 \quad \rightarrow \quad -5x_2^+ + 5x_2^-$

$-4x_2 \quad \rightarrow \quad -4x_2^+ + 4x_2^-$

4

**Add $s_1$ to the $\leq$ constraint and subtract $s_2$ from the $\geq$ constraint**

| | |
|---|---|
| Maximum | $Z = 8x_1 - 2x_2^+ + 2x_2^- + 0s_1 + 0s_2$ |
| S.to | $2x_1 + x_2^+ - x_2^- + s_1 = 12$ |
| | $x_1 - 5x_2^+ + 5x_2^- - s_2 = 6$ |
| | $-4x_2^+ + 4x_2^- = 7$ |
| | $x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$ |

In the answer key: two marks for the methods, two for the objective function, and one for each of the other four lines.

## 05 Basic solutions: from the graph to algebra

After the standard form we have $n$ variables and $m$ equations, with more variables than equations. To solve:

1. Choose n − m variables and set them equal to zero (nonbasic)
2. Solve the m equations for the remaining m variables (basic)
3. The result is a basic solution
4. If all its values are ≥ 0 it is a Basic Feasible Solution (BFS)

| Term | Meaning |
| --- | --- |
| **Nonbasic variables** | The `n − m` variables that we set to zero. |
| **Basic variables** | The `m` variables that we solve for. |
| **Basic solution** | The unique solution of the system after setting the nonbasic variables to zero. |
| **Basic Feasible Solution** | A basic solution with no negative values. Every BFS corresponds to a corner of the feasible region. |

## The maximum number of basic solutions

$$C(n, m) = \frac{n!}{m!\,(n-m)!}$$

$n = 4, m = 2$: $\quad C(4, 2) = \frac{4!}{2! \cdot 2!} = \frac{24}{4} = 6$

**Appeared in the midterm (471)**

True/false items, both true:

> A basic feasible solution is a feasible solution to LPP which is also the basic solution. (T)
> A basic feasible solution is called degenerate if value of at least one basic variable is zero. (T)

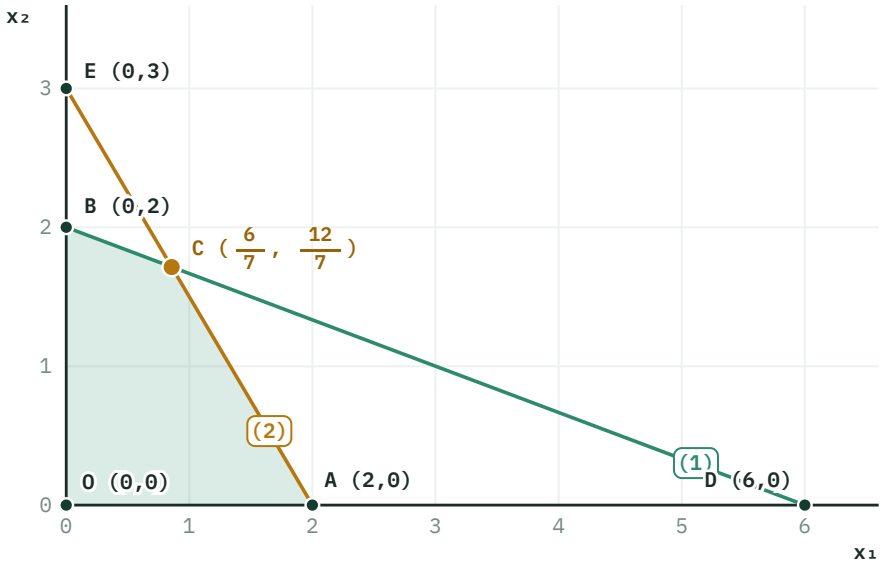
## 06 Example 3.2: all the basic solutions

| |
|---|
| Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$ |
| subject to $x_1 + 3x_2 + s_1 = 6$ |
| $3x_1 + 2x_2 + s_2 = 6$ |
| $x_1, x_2, s_1, s_2 \geq 0$ |

Here $n = 4$ and $m = 2$, so we set two variables to zero each time and solve for the rest. There are 6 cases:

| Nonbasic (= 0) | Basic | Basic solution | Feasible? | z | Corner |
|---|---|---|---|---|---|
| $s_1, s_2$ | $x_1, x_2$ | $(\frac{6}{7}, \frac{12}{7}, 0, 0)$ | Yes | $\frac{48}{7}$ | C |
| $x_2, s_2$ | $x_1, s_1$ | $(2, 0, 4, 0)$ | Yes | 4 | A |
| $x_2, s_1$ | $x_1, s_2$ | $(6, 0, 0, -12)$ | No | — | D |
| $x_1, s_2$ | $x_2, s_1$ | $(0, 3, -3, 0)$ | No | — | E |
| $x_1, s_1$ | $x_2, s_2$ | $(0, 2, 0, 2)$ | Yes | 6 | B |
| $x_1, x_2$ | $s_1, s_2$ | $(0, 0, 6, 6)$ | Yes | 0 | O |

The solution is written in the order $(x_1, x_2, s_1, s_2)$. One row as an example: set $x_1$ and $s_1$ to zero, so the first equation becomes $3x_2 = 6$ giving $x_2 = 2$, and the second becomes $2(2) + s_2 = 6$ giving $s_2 = 2$.




**Optimum** $z = \frac{48}{7}$ **at** $x_1 = \frac{6}{7}$, $x_2 = \frac{12}{7}$ **(corner C)**

The two infeasible solutions, D and E, are on the graph but outside the shaded region. The four feasible ones are exactly the corners of the region.

## 07 Example 3.3: with an unrestricted variable

$$\text{Maximize } z = x_1^+ - x_1^- + 3x_2 + 0s_1 + 0s_2$$

$$\text{subject to} \quad x_1^+ - x_1^- + x_2 + s_1 = 2$$

$$-x_1^+ + x_1^- + x_2 + s_2 = 4$$

$$x_1^+,\ x_1^-,\ x_2,\ s_1,\ s_2 \geq 0$$

Here `n = 5` and `m = 2`, so the maximum is `C(5, 2) = 10`. One case has no solution, so the actual number is 9. The number of basic solutions can be less than the maximum.

| Nonbasic (= 0) | Basic | Basic solution | Feasible? | z |
|---|---|---|---|---|
| $x_2, s_1, s_2$ | $x_1^+, x_1^-$ | no solution | – | – |
| $x_1^-, s_1, s_2$ | $x_1^+, x_2$ | (−1, 0, 3, 0, 0) | No | – |
| $x_1^-, x_2, s_2$ | $x_1^+, s_1$ | (−4, 0, 0, 6, 0) | No | – |
| $x_1^-, x_2, s_1$ | $x_1^+, s_2$ | (2, 0, 0, 0, 6) | Yes | 2 |
| $x_1^+, s_1, s_2$ | $x_1^-, x_2$ | (0, 1, 3, 0, 0) | Yes | **8** |
| $x_1^+, x_2, s_2$ | $x_1^-, s_1$ | (0, 4, 0, 6, 0) | Yes | −4 |
| $x_1^+, x_2, s_1$ | $x_1^-, s_2$ | (0, −2, 0, 0, 6) | No | – |
| $x_1^+, x_1^-, s_2$ | $x_2, s_1$ | (0, 0, 4, −2, 0) | No | – |
| $x_1^+, x_1^-, s_1$ | $x_2, s_2$ | (0, 0, 2, 0, 2) | Yes | 6 |
| $x_1^+, x_1^-, x_2$ | $s_1, s_2$ | (0, 0, 0, 2, 4) | Yes | 0 |

**Optimum $z = 8$ at $x_1^- = 1$, $x_2 = 3$ so $x_1 = 0 - 1 = -1$, $x_2 = 3$**

The slide's table marks the solution `(0, −2, 0, 0, 6)` as feasible, but $x_1^- = -2$ is negative, so it is not feasible. The table here is corrected.

## 08 Example 3.5: absolute values

Convert the following optimization problem into a LPP:

Maximum $z = \max\{ |2x_1 + 3x_2|, |3x_1 - 7x_2| \}$

Subject to: $x_1 + x_2 \le 2$, $-x_1 + x_2 \le 4$, $x_1, x_2 \ge 0$

The idea: call the larger value `y`. Since `|a| ≤ y` means `−y ≤ a ≤ y`, each absolute value gives two linear constraints:

| | |
|---|---|
| Maximize $Z = y$ | |
| subject to | $x_1 + x_2 \leq 2$ |
| | $-x_1 + x_2 \leq 4$ |
| | $2x_1 + 3x_2 - y \leq 0$ |
| | $-2x_1 - 3x_2 - y \leq 0$ |
| | $3x_1 - 7x_2 - y \leq 0$ |
| | $-3x_1 + 7x_2 - y \leq 0$ |
| | $x_1, x_2, y \geq 0$ |

**A note on this example**

In the exam, write the solution as above, since it is the slide's form. But note a mathematical flaw: the last four constraints all say $y$ is at least both values, so they bound $y$ only from below and nothing bounds it from above. If the objective is maximize, $y$ can grow without limit and every constraint still holds, so the problem is unbounded. This conversion works correctly when the objective is minimize, because the smallest $y$ that covers both values is exactly the larger of them.

## 09 Common mistakes

| Mistake | Correct |
|---|---|
| Adding a slack to a $\geq$ constraint | A surplus is subtracted from a $\geq$ constraint: $-\ s_2$. |
| Adding a variable to an $=$ constraint | It gets no slack and no surplus. |
| Leaving a negative right-hand side | $4x_2 = -7$ becomes $-4x_2 = 7$. |
| Substituting the unrestricted variable in the constraints but not in the objective | It is substituted everywhere it appears. |
| Losing signs with a negative coefficient | $-5x_2$ becomes $-5x_2^+ + 5x_2^-$. |
| Forgetting the last line | $x_1,\ x_2^+,\ x_2^-,\ s_1,\ s_2 \geq 0$ |

## 10 Test yourself

1. What does the standard form mean? (two requirements)
2. What is the difference between a slack and a surplus variable?
3. What is the maximum number of basic solutions when $n = 5$ and $m = 3$?
4. Write in standard form: $\text{Min } Z = 3x_1 + 2x_2$ subject to $x_1 + x_2 \leq 9$, $2x_1 - x_2 \geq 4$, $3x_2 = -6$, with $x_1 \geq 0$ and $x_2$ unrestricted.

## Answers

| # | Answer |
|---|---|
| 1 | All decision variables are nonnegative, and all constraints are equations with nonnegative RHS. |
| 2 | A slack is added to a `≤` constraint, and a surplus is subtracted from a `≥` constraint. |
| 3 | $C(5,\ 3) = \frac{5!}{3!\ \cdot\ 2!} = 10$ |
| 4 | Min $Z = 3x_1 + 2x_2^+ - 2x_2^- + 0s_1 + 0s_2$<br>$x_1 + x_2^+ - x_2^- + s_1 = 9$<br>$2x_1 - x_2^+ + x_2^- - s_2 = 4$<br>$-3x_2^+ + 3x_2^- = 6$<br>$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$ |